

Top 5 Customers

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,

SUM(i.Total) AS TotalSales

FROM Customer c

INNER JOIN Invoice i

ON c.CustomerId = i.CustomerId

GROUP BY CustomerName

ORDER BY TotalSales DESC

LIMIT 5

"

)

in

Source

Top 5 Genres

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

g.Name AS Genre,

SUM(il.UnitPrice * il.Quantity) AS TotalSales

FROM Genre g

INNER JOIN Track t

ON g.GenreId = t.GenreId

INNER JOIN InvoiceLine il

ON t.TrackId = il.TrackId

GROUP BY g.Name

ORDER BY TotalSales DESC

LIMIT 5

"

)

in

Source

Top 5 Artists

let

```
Source = Odbc.Query(  
  "dsn=Route Project",  
  "  
  SELECT  
    ar.Name AS ArtistName,  
    SUM(il.UnitPrice * il.Quantity) AS Revenue  
  FROM Artist ar  
  INNER JOIN Album al  
    ON ar.ArtistId = al.ArtistId  
  INNER JOIN Track t  
    ON al.AlbumId = t.AlbumId  
  INNER JOIN InvoiceLine il  
    ON t.TrackId = il.TrackId  
  GROUP BY ar.Name  
  ORDER BY Revenue DESC  
  LIMIT 5  
  "  
)
```

in

Source

Top 5 Albums by Revenue

let

```
Source = Odbc.Query(  
  "dsn=Route Project",  
  "  
  SELECT  
    al.Title AS AlbumTitle,  
    SUM(il.UnitPrice * il.Quantity) AS Revenue  
  FROM Album al  
  INNER JOIN Track t  
    ON al.AlbumId = t.AlbumId  
  INNER JOIN InvoiceLine il  
    ON t.TrackId = il.TrackId  
  GROUP BY al.Title  
  ORDER BY Revenue DESC  
  LIMIT 10  
  "  
)
```

in

Source

Tracks Count by Genre

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        g.Name AS Genre,  
        COUNT(t.TrackId) AS TotalTracks  
    FROM Genre g  
    INNER JOIN Track t  
        ON g.GenreId = t.GenreId  
    GROUP BY g.Name  
    ORDER BY TotalTracks DESC  
    "  
    )
```

in

Source

Average Duration by Genre

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        g.Name AS Genre,  
        ROUND(AVG(t.Milliseconds)/60000,2) AS AvgMinutes  
    FROM Genre g  
    INNER JOIN Track t  
        ON g.GenreId = t.GenreId  
    GROUP BY g.Name  
    ORDER BY AvgMinutes DESC  
    "  
    )
```

in

Source

Top 5 Most Purchased Tracks

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        t.Name AS TrackName,  
        SUM(il.Quantity) AS TotalSold  
    FROM Track t  
    INNER JOIN InvoiceLine il  
        ON t.TrackId = il.TrackId  
    GROUP BY t.Name  
    ORDER BY TotalSold DESC  
    LIMIT 10  
    "  
    )
```

in

Source

Revenue by City

let

```
Source = Odbc.Query(  
  "dsn=Route Project",  
  "  
  SELECT  
    c.City,  
    SUM(i.Total) AS Revenue  
  FROM Customer c  
  INNER JOIN Invoice i  
    ON c.CustomerId = i.CustomerId  
  GROUP BY c.City  
  ORDER BY Revenue DESC
```

Limit 5

```
"  
  
)
```

in

Source

Top 5 Countries by Number of Customers

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

Country,

COUNT(CustomerId) AS CustomersCount

FROM Customer

GROUP BY Country

ORDER BY CustomersCount DESC

LIMIT 5

"

)

in

Source

Customers Above Average Spending (Subquery)

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        CustomerRevenue.CustomerName,  
        CustomerRevenue.Revenue  
    FROM  
    (  
        SELECT  
            CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,  
            SUM(i.Total) AS Revenue  
        FROM Customer c  
        INNER JOIN Invoice i  
            ON c.CustomerId=i.CustomerId  
        GROUP BY CustomerName  
    ) CustomerRevenue  
  
    WHERE Revenue >  
    (  
        SELECT AVG(CustomerTotal)  
    FROM  
    (  
        SELECT
```

```
        SUM(i.Total) AS CustomerTotal
    FROM Invoice i
    GROUP BY i.CustomerId
    ) AvgTable
)

ORDER BY Revenue DESC

limit 5
"
)
in
Source
```

Visual 3: Customer Purchase Frequency

let

```
Source = Odbc.Query(  
  "dsn=Route Project",  
  "  
  SELECT  
    CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,  
    COUNT(i.InvoiceId) AS NumberOfPurchases  
  FROM Customer c  
  INNER JOIN Invoice i  
    ON c.CustomerId = i.CustomerId  
  GROUP BY CustomerName  
  ORDER BY NumberOfPurchases DESC  
  LIMIT 5  
  "  
)
```

in

Source

Customer Lifetime Value (CLV)

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,

COUNT(i.InvoiceId) AS TotalOrders,

SUM(i.Total) AS LifetimeValue

FROM Customer c

INNER JOIN Invoice i

ON c.CustomerId = i.CustomerId

GROUP BY CustomerName

ORDER BY LifetimeValue DESC

LIMIT 10

"

)

in

Source

Average Order Value by Country

let

```
Source = Odbc.Query(  
  "dsn=Route Project",  
  "  
  SELECT  
    c.Country,  
    ROUND(AVG(i.Total),2) AS AvgOrderValue  
  FROM Customer c  
  INNER JOIN Invoice i  
    ON c.CustomerId = i.CustomerId  
  GROUP BY c.Country  
  ORDER BY AvgOrderValue DESC  
  LIMIT 5  
  "  
)
```

in

Source

Monthly Revenue Trend

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        YEAR(InvoiceDate) AS SalesYear,  
        MONTH(InvoiceDate) AS SalesMonth,  
        SUM(Total) AS Revenue  
    FROM Invoice  
    GROUP BY  
        YEAR(InvoiceDate),  
        MONTH(InvoiceDate)  
    ORDER BY  
        SalesYear,  
        SalesMonth  
    "  
    )
```

in

Source

Running Revenue Total

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

InvoiceDate,

SUM(Total) AS Revenue,

SUM(SUM(Total))

OVER(

ORDER BY InvoiceDate

) AS RunningRevenue

FROM Invoice

GROUP BY InvoiceDate

ORDER BY InvoiceDate

"

)

in

Source

Revenue Rank by Country

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

BillingCountry,

SUM(Total) AS Revenue,

RANK() OVER(

ORDER BY SUM(Total) DESC

) AS RevenueRank

FROM Invoice

GROUP BY BillingCountry

"

)

in

Source

Top 5 Months by Revenue

let

Source = Odbc.Query(

"dsn=Route Project",

"

SELECT

DATE_FORMAT(InvoiceDate,'%Y-%m') AS SalesMonth,

SUM(Total) AS Revenue

FROM Invoice

GROUP BY SalesMonth

ORDER BY Revenue DESC

LIMIT 10

"

)

in

Source

Sales by Media Type

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        mt.Name AS MediaType,  
        SUM(il.UnitPrice * il.Quantity) AS Revenue  
    FROM MediaType mt  
    INNER JOIN Track t  
        ON mt.MediaTypeId = t.MediaTypeId  
    INNER JOIN InvoiceLine il  
        ON t.TrackId = il.TrackId  
    GROUP BY mt.Name  
    ORDER BY Revenue DESC  
    "  
    )
```

in

Source

Revenue by Employee

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,  
        SUM(i.Total) AS Revenue  
    FROM Employee e  
    INNER JOIN Customer c  
        ON e.EmployeeId = c.SupportRepId  
    INNER JOIN Invoice i  
        ON c.CustomerId = i.CustomerId  
    GROUP BY EmployeeName  
    ORDER BY Revenue DESC  
    "  
    )
```

in

Source

Revenue by Customer Support Representative

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,  
        SUM(i.Total) AS Revenue  
    FROM Employee e  
    INNER JOIN Customer c  
        ON e.EmployeeId = c.SupportRepId  
    INNER JOIN Invoice i  
        ON c.CustomerId = i.CustomerId  
    GROUP BY EmployeeName  
    ORDER BY Revenue DESC  
    "  
    )
```

in

Source

Customers Managed by Employee

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,  
        COUNT(c.CustomerId) AS CustomersCount  
    FROM Employee e  
    INNER JOIN Customer c  
        ON e.EmployeeId = c.SupportRepId  
    GROUP BY EmployeeName  
    ORDER BY CustomersCount DESC  
    "  
    )
```

in

Source

Revenue per Customer by Employee

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        CONCAT(e.FirstName, ' ', e.LastName) AS EmployeeName,  
        ROUND(  
            SUM(i.Total) / COUNT(DISTINCT c.CustomerId),  
            2  
        ) AS RevenuePerCustomer  
    FROM Employee e  
    INNER JOIN Customer c  
        ON e.EmployeeId = c.SupportRepId  
    INNER JOIN Invoice i  
        ON c.CustomerId = i.CustomerId  
    GROUP BY EmployeeName  
    ORDER BY RevenuePerCustomer DESC  
    "  
    )
```

in

Source

Tracks by Media Type

let

```
Source = Odbc.Query(  
    "dsn=Route Project",  
    "  
    SELECT  
        mt.Name AS MediaType,  
        COUNT(t.TrackId) AS TracksCount  
    FROM MediaType mt  
    INNER JOIN Track t  
        ON mt.MediaTypeId = t.MediaTypeId  
    GROUP BY mt.Name  
    ORDER BY TracksCount DESC  
    "  
    )
```

in

Source

Top 5 Companies by Revenue

let

```
Source = Odbc.Query(  
  "dsn=Route Project",  
  "  
  SELECT  
    c.Company,  
    ROUND(SUM(i.Total),2) AS Revenue  
  FROM Customer c  
  INNER JOIN Invoice i  
    ON c.CustomerId = i.CustomerId  
  WHERE c.Company IS NOT NULL  
  GROUP BY c.Company  
  ORDER BY Revenue DESC  
  LIMIT 5  
  "  
)
```

in

Source